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Relativistic Effects in Chemistry: More Common Than You Thought

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Abstract

Relativistic effects can strongly influence the chemical and physical properties of heavy elements and their compounds. This influence has been noted in inorganic chemistry textbooks for a couple of decades. This review provides both traditional and new examples of these effects, including the special properties of gold, lead-acid and mercury batteries, the shapes of gold and thallium clusters, heavy-atom shifts in NMR, topological insulators, and certain specific heats.

1. INTRODUCTION

Relativistic effects are important for fast-moving particles. Because the average speeds of valence electrons are low, it was originally thought [in fact by Dirac (1) himself] that relativity then was unimportant. It has now been known for a while that relativistic effects can strongly influence many chemical properties of the heavier elements (2–5). Well-confirmed examples include the yellow color, nobility, and trivalency of gold and the large effects on the bond lengths. A probable, but not explicitly demonstrated, consequence is the liquidity of mercury at room temperature. A recent example is the lead-acid battery that derives most of its voltage from relativistic effects.

In a broad sense, the differences between the sixth period (Cs through Rn) and the preceding fifth period (Rb through Xe) largely result from relativistic effects and the lanthanide contraction (the traditional explanation). This information has been noted in chemistry textbooks for a couple of decades now.

In this review I find it useful to repeat key arguments and mention the latest examples and detailed explanations and confirmations. The fundamental aspects (mainly the next physical level of quantum electrodynamics) are discussed in a companion review (6). A new Periodic Table (PT) up to $Z = 172$ has been suggested in Reference 7. Since the publication of Reference 5 and its supplement (8), other reviews on relativity in chemistry have appeared, including those by Balasubramanian (9) and Kaltsoyannis (10) (for main-group chemistry, see 11).

2. FUNDAMENTALS

2.1. Simple Estimates and Textbooks

Among the most important consequences of relativistic quantum chemistry are the simple explanations it provides for teaching and understanding the chemistry of the heavier elements.

2.1.1. A simple argument. A simple argument (probably first published in Reference 2) that makes relativistic effects plausible is the following.

The inner electrons move fast in heavy elements. For the innermost, $1s$ shell, the average radial velocity is for a nonrelativistic, hydrogenlike approximation

$$\langle v_r \rangle_{1s} = Z \quad (1)$$

$$= 80 \text{ for Hg} \quad (2)$$

in atomic units, where the speed of light, c , is $= 137.035999679(94)$ (year 2008 standard value). This leads to a mass increase,

$$m = \gamma m_0 \quad (3)$$

$$= m_0 / \sqrt{1 - (v/c)^2}. \quad (4)$$

The increased mass gives a smaller Bohr radius

$$a_0 = \hbar^2 / m e^2. \quad (5)$$

This yields a relativistic contraction and stabilization of all s and most p orbitals of many-electron atoms. The nonrelativistic binding energy is $E_n = -\frac{Z^2}{2n^2}$, and the first relativistic correction to it will be of order $E_n^{\text{rel}} = -\frac{Z^4}{2n^3 c^2}$. For hydrogenlike atoms, an exact solution of the Dirac equation shows that the higher s and p states are percentally as strongly relativistic as their inner counterparts. Moreover, because of the stronger screening of the nuclear attraction by the contracted s and p

UNDERSTANDING THE SPIN-ORBIT COUPLING

The SO term splits atomic $p, d, \dots$ levels into the pairs $(p_{3/2}, p_{1/2}), (d_{5/2}, d_{3/2})$, and so on, corresponding to a total angular momentum $j = l \pm \frac{1}{2}$.

Section 2.1.1 demonstrates that a hydrogenlike atom can have relativistic energy contributions of the order $c^{-2}Z^4$ a.u. The SO coupling of the electron spin magnetic moment $\mu = g_e s$ with the orbital angular momentum $\mathbf{l}$ for quantum number $l > 0$ has the same order of magnitude. How do we see that, and why is it a relativistic effect? Two useful textbooks are Moss (12) and Atkins & Friedman (13, pp. 215–17, 238).

A particle moving with velocity $\mathbf{v}$ in electric field $\mathbf{E}$ will see a magnetic field

$$\mathbf{B} = \frac{1}{c^2} \mathbf{E} \times \mathbf{v}. \quad (\text{S1})$$

This is a relativistic effect, an element in a Lorentz transformation. [This is to the lowest level. The full expression is of type $B_y = \gamma(B'_y - vE'_z/c^2)$, where γ is defined in Equation 4 (see Reference 12, p. 69).] We also obtain c^{-2} from this equation. In a hydrogenlike atom, the typical v grows like Z , and the typical E grows like Z^3 (one power from the nuclear charge, two powers from the typical r^{-2}), so we obtain the desired $c^{-2}Z^4$ interaction.

For a spherically symmetrical potential $\phi(r)$,

$$\mathbf{E} = -\frac{\mathbf{r}}{r} \phi', \quad (\text{S2})$$

with $\phi' = \frac{d\phi}{dr}$. Hence

$$\mathbf{B} = -\frac{1}{rc^2} \phi' \mathbf{r} \times \mathbf{v}. \quad (\text{S3})$$

As $\mathbf{l} = \mathbf{r} \times \mathbf{p}$, and hence $\mathbf{r} \times \mathbf{v} = \frac{1}{m_e} \mathbf{l}$, we get the Hamiltonian

$$b_{\text{SO}} = -\mu_e \cdot \mathbf{B} = \frac{1}{m_e r c^2} \phi' \mu \cdot \mathbf{l} = \frac{e}{m_e^2 r c^2} \phi' \mathbf{s} \cdot \mathbf{l}, \quad (\text{S4})$$

which must still be divided by 2, the celebrated Thomas factor of two, because of a further Lorentz transformation to the electron rest frame (for a simple derivation, see 12, pp. 81–84).

As discussed in Section 2.3.1, the hydrogenlike Z^4 trend is changed to an approximate Z^2 one for both scalar and SO relativistic effects for the valence electrons of analogous many-electron systems.

shells, one obtains in many-electron atoms a relativistic expansion and destabilization of d and f shells. These effects are large enough to substantially contribute to the chemical differences between periods 5 (Rb through Xe) and 6 (Cs through Rn) of the PT. Both these direct and indirect effects and the spin-orbit (SO) splitting increase for valence shells down a given column roughly as Z^2 . Here Z is the full nuclear charge. In hydrogenlike systems, one would have the Z^4 trend (see Understanding the Spin-Orbit Coupling, sidebar above).

2.1.2. The entry into chemistry textbooks. Some chemistry textbooks that introduce relativity ideas are listed in **Table 1**.

Some chemical trends that can then be qualitatively explained include the following:

- Why is gold noble? This is owing to its larger $6s$ binding energy. Moreover, gold is tri- or pentavalent because of its smaller $5d$ binding energy (for explicit calculations, see Reference 14). Moreover, its yellow color is caused by the smaller gap from the filled $5d$ shell to the half-filled $6s$ band (see Section 3.1 below for a full discussion).

Table 1 Some inorganic chemistry textbooks introducing relativity ideas*

Authors	Year (year of edition ideas first introduced)	Reference
Wulfsberg	1991 (1987)	17 (see chapter 1-8 and also pp. 175, 260, 1084)
Cotton et al.	1999 (1988)	18 (see chapter 16.13)
Mackay et al.	1996 (1989)	19
Huheey et al.	1993	20 (see pp. 579, 879–80)
Norman ^a	1997 (1994)	21 (see p. 30)
Hollemann et al.	2007 (1995)	22 (see chapter 2.1.4, pp. 338–40)
Greenwood & Earnshaw	1997	23 (see pp. 599, 1180, 1266, 1274)
Mingos	1998	24 (see pp. 26, 367)

*See author comment in Reference 22.

^aA British school textbook.

- Why are aurides [Au(–I) compounds (15)] so common? This results from the larger $6s$ binding energy, seen by the $6s$ hole, which is a reflection of a higher electron affinity (EA). Even the isoelectronic Pt^{2-} compound $(\text{Ba}^{2+})_2(\text{Pt}^{2-})(2e^-)$ has been made (16).
- Additionally, CsAu is a relativistic semiconductor, and $\text{CsAu}(\text{NR})$ would be a metal (see Reference 5, p. 578).
- Why is mercury liquid? It is probably because the filled $6s^2$ shell is now more stable. However, explicit proof is still missing.
- There is also the existence of atomic ground-state changes, such as $\text{Mo } 4d^5 5s^1$ but $\text{W } 5d^4 6s^2$ (s down, d up). Changes also occur for the main oxidation state from $\text{Sn}(\text{IV})$ to $\text{Pb}(\text{II})$ (at least partly because $6s$ was stabilized). With regard to diatomic Tl_2 , it has a small dissociation energy, resulting from the larger SO stabilization of the $6p^1$ atoms than that of the molecule. Finally, there is the existence of monovalent $\text{Bi}(\text{I})$ compounds, caused by the SO stabilization of the filled $6p^* = 6p_{1/2}$ subshell.

2.2. The Dirac-Coulomb-Breit Electronic Hamiltonian

A good basis for a quantitative treatment is the DCB (Dirac-Coulomb-Breit) Hamiltonian. For electrons in nuclear potential V_n , it can be written as

$$H = \sum_i b_i + \sum_{i < j} b_{ij}. \quad (6)$$

The one-particle Dirac Hamiltonian

$$b_i = c \boldsymbol{\alpha} \cdot \mathbf{p} + \beta c^2 + V_n, \quad \mathbf{p} = -i\nabla. \quad (7)$$

The two-particle Hamiltonian

$$b_{ij} = b_C + b_B, \quad b_C = 1/r_{ij}, \quad (8)$$

where

$$b_B = -\frac{1}{2r_{ij}} [\boldsymbol{\alpha}_i \cdot \boldsymbol{\alpha}_j + (\boldsymbol{\alpha}_i \cdot \mathbf{r}_{ij})(\boldsymbol{\alpha}_j \cdot \mathbf{r}_{ij})/r_{ij}^2]. \quad (9)$$

For b_B , there are alternative, frequency-dependent forms (see, e.g., 25). In the Coulomb gauge used for a magnetic vector potential $\mathbf{A}$, one sets $\nabla \cdot \mathbf{A} = 0$. Then the electron-electron interactions can be taken as instantaneous. In correlated calculations (beyond single-Slater-determinant,

self-consistent-field ones), electron-like projection operators, P , should be added:

$$h_{ij}^{\text{eff}} = Ph_{ij}P. \quad (10)$$

This is also called the no-virtual-pair approximation.

This DCB description is good but is not physically complete. The Araki term and the quantum electrodynamic terms are discussed in a companion review (6). For the heavier elements ($Z > 50$), they are of the order of -1% of the Dirac-level relativistic effects. Compared with them, the DCB Hamiltonian is 101% right.

2.3. Some Interpretative Issues

As mentioned in Section 1, nature is sometimes a little more complicated than even the top scientists initially imagined. Moreover, sometimes the same phenomenon can be analyzed from different vantage points.

2.3.1. Direct and indirect relativistic effects. How do the relativistic effects on the valence orbitals arise? Analyzing the direct effects on the valence electrons themselves as a function of the distance from the nucleus, r , Schwarz et al. (26) found that a large part of the relativistic changes comes from the innermost half-wave (i.e., the $1s$ domain for a valence ns orbital, and so on). The same conclusion was reached earlier by Dehmer (27) for the SO splitting: For a valence np shell, most of the SO arises from the innermost, $2p$ -like domain. Because the part of the total norm in this first half-wave decreases roughly as Z^{-2} , the hydrogen-like Z^4 increase is cut to an approximate Z^2 one in the valence shell for similar systems, down a column of the PT, with Z the total, unscreened nuclear charge. This approach to analyzing relativistic effects could also be seen as a way to fix the correct phase and amplitude of the oscillating, radial one-electron wave functions at the outer limit of the core, qualitatively explaining the effectiveness of pseudopotential (effective core potential) methods.

The predominant relativistic effects on s and p shells are direct ones on the valence electron dynamics. There also are indirect contributions from the relativistic changes of the other orbitals (for an example, see Section 3.2).

2.3.2. A word on the available methods of calculation. In the present review, we describe only selected chemical examples. The methods used have been recently described by Schwerdtfeger (28, 29), Hess (30), Hirao & Ishikawa (31), Dylla & Faegri (32), Grant (33), Reiher & Wolf (34), and Barysz & Ishikawa (35). These methods range from fully relativistic (four-component Dirac) ones to transformed Hamiltonians, such as the exact two-component approach (36). A successful approach involves pseudopotentials (effective core potentials) (37, 38). They can be used with common codes such as Gaussian, Molpro, Molcas, or Turbomole. Both density functional theory (DFT) and its counterpart wave-function theory (WFT) are in common use. Electron correlation can be handled in the latter case up to the coupled-cluster level with single, double, and perturbative triple excitations, CCSD(T), and beyond.

3. SOME CLASSICAL EXAMPLES

3.1. The Yellow Color of Gold

As noted in References 4 and 5, comparative relativistic/nonrelativistic band-structure calculations on gold have been available since the 1960s, and these show that the excitation energies from the

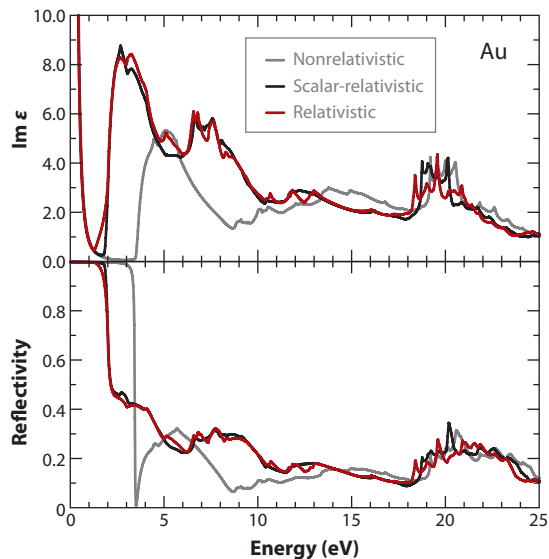


Figure 1

Calculated nonrelativistic, scalar relativistic, and relativistic dielectric constants for bulk metallic gold. Note the (*gray-to-red*) relativistic shift from 3.5 to 2 eV in both curves. The upper and lower panels give the imaginary and real parts, respectively. Figure reprinted from Reference 41 with permission, copyright Institute of Physics.

top of the $5d$ band to the Fermi level, in the half-filled $6s$ band, lie in the middle of the visible energy range when relativistic effects were included. Without relativistic effects, that excitation energy would be much larger, in the UV. This was brought in contact with immediate visual impressions in Reference 4, although not much novelty can be claimed for the word “yellow,” introduced there.

Still missing were explicit calculations of the dielectric constants for gold. They have been reported quite recently by Romaniello & de Boeij (39, 40). As seen in **Figure 1** (from a later confirmation), the onset of the optical absorption in the middle of the visible, near 2 eV, is well reproduced. In a corresponding nonrelativistic calculation, that threshold is moved to approximately 3.6 eV, in the UV. Thus nonrelativistic gold is white, like silver, and the yellow color of gold indeed comes from relativity.

These were still bulk, not surface, calculations. The ensuing differences are estimated to be small (P.L. de Boeij, private communication, 2005). In a later study, Glantschnig & Ambrosch-Draxl (41) emphasized the SO aspects on gold and several other metals, up to the far UV.

Do other relativistic colors exist? The violet color of pentaphenyl bismuth, BiPh_5 , and the yellow color of hexachloroplumbate(IV), PbCl_6^{2-} , have been attributed to the relativistic stabilization of an a_1^* lowest unoccupied molecular orbital (LUMO). The starting point of the electronic excitation was a ligand orbital, but the empty, antibonding, upper state came down owing to its heavy-metal $6s$ character (42). The corresponding Sb and Sn compounds are colorless (**Figure 2**). In the case of $\text{Pb}(\text{NO}_2)_2$, the color is attributed to a singlet-triplet transition of the nitrite, induced by the SO coupling of the Pb center.

Another simple example on the relativistic stabilization of an originally empty shell, the $8s$, is the calculated EA of 0.064(2) eV for the noble gas E118 (43). The size of the anion was not discussed, but by the uncertainty principle, it probably is the largest monoatomic ion known to humankind.

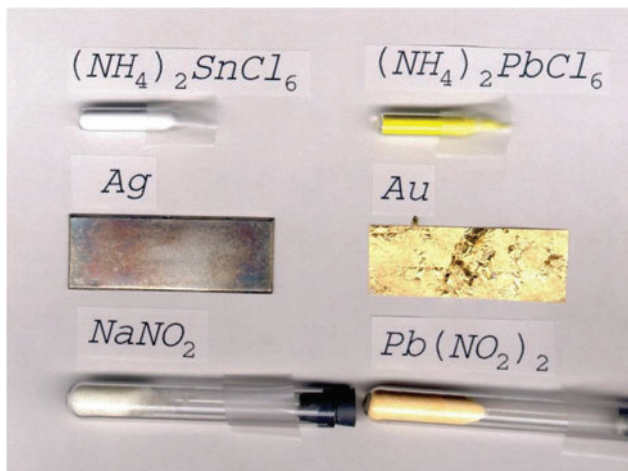


Figure 2

Three sets of light- and heavy-element systems in which the yellow colors of the latter are attributed to relativistic effects. Figure taken from Reference 150.

3.2. The Gold Maximum of Relativistic Effects

When analyzing earlier atomic calculations by Desclaux, Pyykkö & Desclaux (4) found that the radial contraction $\langle r \rangle_R / \langle r \rangle_{NR}$ for a $6s$ orbital had local minima in period 6 (Cs through Rn) of the PT for groups 1 and 18 and a pronounced maximum at the gold atom in group 11. Similar local maxima of relativistic effects occurred at Cu and Ag in periods 4 and 5, respectively.

The underlying reasons were analyzed by Autschbach et al. (44). When passing in the PT from $_{70}\text{Yb}$ to $_{80}\text{Hg}$, or from group 2 to 12 ($g = 2 \rightarrow g = 12$), the two common electron configurations are $d^{g-2}s^2$ and $d^{g-1}s^1$. Both the $6s$ -electron binding energy itself and its relativistic increase grow along the series, but the two electron configurations follow separate curves. Defining for a property x a relative change

$$(\Delta_{\text{rel}}x)/x = (x_R - x_{NR})/x_R = \gamma_s(nl) (Z/c)^2, \quad (11)$$

Autschbach et al. found rather similar trends for the ns orbital energies, ϵ , of the $3d$, $4d$, and $5d$ metal atoms, with the prefactor γ_ϵ increasing from approximately 0.2 to approximately 0.4–0.6 for the s^2 configuration. For the group-11 s^1 configuration, γ_ϵ rose to approximately 0.5–0.7. Thus the partial screening from the inner $(n-1)d$ shell increases both $|\epsilon_{ns}|$ and its $\gamma_\epsilon(ns)$ factor. This leads to the gold maximum at group 11. We also note that the actinides have large $\gamma_\epsilon(6s)$ values (44). These $\gamma_\epsilon(6s)$ values can be much larger for the ns valence orbitals of neutral atoms than for the ns orbital of a one-electron atom with the same Z .

For the $(n-1)d$ orbitals in a relativistic all-electron calculation, the γ_ϵ factors are negative. This is the above-mentioned indirect destabilization effect due to the stabilization and contraction of the s and p orbitals. In a one-electron atom, $\gamma_\epsilon[(n-1)d]$ would still be positive.

We can be more specific and ask whether the $\gamma_\epsilon(6s)$ increase with an increasing number of $5d$ electrons from Lu to Hg, because the particular effective potential yields a stronger direct relativistic effect, or is there a self-consistent, indirect effect in which the expansion of $5d$ enhances the contraction of $6s$? Schwarz et al. (26, figure 4) demonstrated that the answer for gold is mostly direct, a conclusion already reached by Rose et al. (45).

3.3. High Oxidation States, High Electron Affinities

It was suggested in Reference 2 that the *5d* metals do have higher oxidation states than their *4d* analogs because of the destabilization of their *d* shell. A striking example is the predicted HgF_4 , which is the first Hg(IV) compound (46). Another example is Ir(VIII)O_4 (47). The predicted IrO_4^+ would have the first oxidation state +IX (48). The predicted (49) octahedral UO_6 remains a local minimum (50) but does not have the high charge at the U atom to classify it as a U(XII) compound. Moreover, there are lower-lying alternative peroxide structures. For a review on the high oxidation states, readers are referred to Riedel & Kaupp (51). The *5d* metal hexafluorides WF_6 through AuF_6 have high electron affinities and are extraordinary oxidizers and Lewis acids (52); the SO increased the EA.

3.4. The Spin-Spin Coupling and Heavy-Atom Shifts in NMR

A comprehensive summary of the theory of NMR and electron paramagnetic resonance parameters was published in 2004 (53). The hyperfine operators involved are strong close to the nuclei. However, even the relativistic *s*-state (corresponding to the nonrelativistic Fermi contact) operator gets its main contribution from the *1s*-like domain, not a closer one (54). If the relativistic/nonrelativistic ratio is expressed as a multiplicative correction factor, it is 2.5733 or 3.0795 for the *6s* shell of $Z = 82$ at the H-like (54) or HF level (55), respectively. For a *J* (Pb-Pb) coupling constant, its square gives an enhancement that is close to one power of 10. The latest references on heavy-element spin-spin coupling can be traced back from Zheng & Autschbach (56). For all NMR parameters, readers are referred to Autschbach & Zheng (57), Kutzelnigg & Liu (58), and for all terms at the Breit-Pauli level, readers are referred to Vaara et al. (59) and Manninen et al. (60).

With regard to chemical shifts, the ^{13}C signal in heavy halomethanes suffers an upfield shift, known as a heavy-atom shift. The heavier and more numerous the heavy halogens are, the larger the shift. Nomura et al. (61) attributed this shift to SO effects on the heavy center(s). The spin polarization created by the heavy-atom SO propagates in the molecular electronic system much like the indirect spin-spin coupling (62).

A recent example involves the ^1H shifts of H-ML_n systems in which M is a transition metal. The Buckingham-Stephens model has to be completed by SO contributions (63). An example is shown in Figure 3.

As is well known, the variations of the *g* tensor from the free-electron value are directly induced by the SO coupling.

3.5. Relativistic Effects on Bond Lengths

In most cases, the effect of relativity on chemical bond lengths, *R*, is a contraction

$$C = R_{\text{NR}} - R_{\text{R}}. \quad (12)$$

For related compounds in the same column of the PT, the contraction again scales as Z^2 :

$$C/\text{pm} = c_Z Z^2. \quad (13)$$

As discussed in Reference 5, c_Z strongly varies as a function of the group in the PT, with a maximum at the coinage metals (group 11), where a $c_Z = 0.0032(7)$ pm was found.

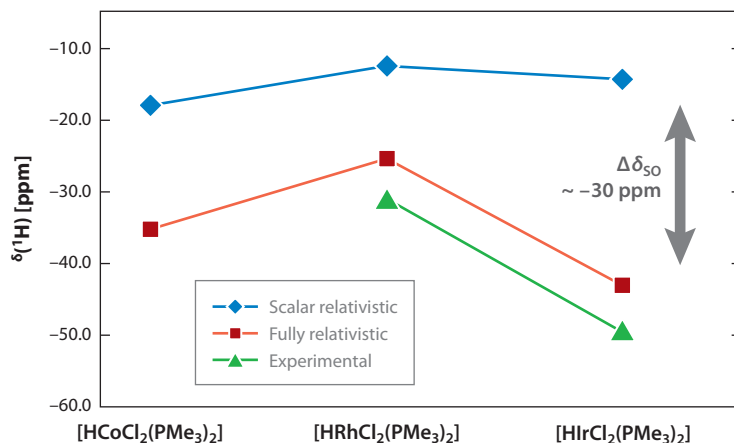


Figure 3

The experimental and calculated ^1H NMR shifts of the (18-electron d^6) complexes $[\text{HMC}\text{Cl}_2(\text{PR}_3)_2]$; $\text{M} = \text{Co}, \text{Rh}, \text{and Ir}$. Note the importance of the spin-orbit (SO) contribution for Rh and Ir. The scalar relativistic (SR) contribution corresponds to the Buckingham-Stephens paramagnetic mechanism. Figure taken from Reference 63 with permission, copyright ACS.

The contraction of bond lengths does not require the contraction of the orbitals, as first found by Ziegler et al. (64) (for further discussion, see References 5 and 65).

3.6. Metallophilicity

An aurophilic or more generally metallophilic attraction means that there is an apparent closed-shell interaction between two or more closed-shell metal ions, such as the $5d^{10}$ Au(I) or the $5d^{10}6s^2$ Tl(I). Recent experimental summaries have been provided by Schmidbaur & Schier (66), Doerrer (67), and Sculfort & Braunstein (68). For recent summaries on the theory, readers are referred to References 69–71. Early semiempirical models were able to reproduce the attraction by $6s-6p-5d$ hybridization. At the wave-function-based *ab initio* level, the largest contribution turns out to be dispersion (van der Waals) forces, with the second largest contribution being virtual charge transfer or ionic interactions.

What is the role of relativity here? Earlier calculations at the lowest possible, MP2 (second-order Møller-Plesset) level demonstrated a relativistic increase. Later studies at higher levels up to CCSD(T), comparing silver [which is essentially nonrelativistic gold (72)] with gold, showed the opposite trend. Thus in the cases studied by O'Grady & Kaltsoyannis (73), the argentophilic attraction was stronger than the aurophilic one. By this evidence, relativistic effects would actually somewhat weaken the group-11 $\text{M(I)}-\text{M(I)}$ interaction.

3.7. Lanthanides and Actinides

The lanthanide contraction of the $\text{Ln}-\text{X}$ bond lengths in LnX_3 molecules from $\text{Ln} = \text{La}$ to $\text{Ln} = \text{Lu}$ is partially a relativistic effect. The latest estimate gives 9%–23% relativity, depending on the system (74). The main valence orbitals of the Ln, forming their covalent bonds, are the $6s$ and the $5d$.

The latest comprehensive treatment of theoretical actinide chemistry has been provided by Kaltsoyannis et al. (75). Readers are also referred to Dolg and colleagues (76, 77) and

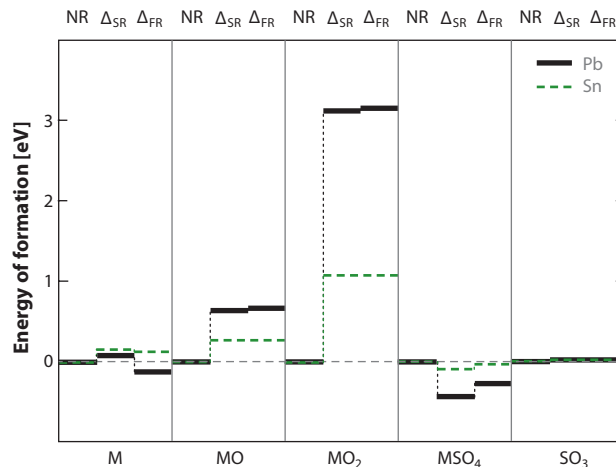


Figure 4

The nonrelativistic (NR), scalar relativistic (SR), and fully relativistic (FR) energy shifts (in electron volts) for the solids involved in the lead-acid-battery reaction (Equation 14). Values for both $M = \text{Sn}$ (green) and $M = \text{Pb}$ (black) are given. Figure reproduced with permission from Reference 80, copyright APS.

Schreckenbach & Shamov (78). The chemical properties of the superheavy elements have notably been calculated by Pershina (79) (see also 7, and references therein).

4. SOME RECENT EXAMPLES

4.1. The Lead-Acid Battery

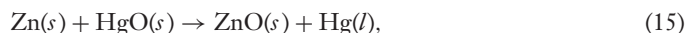
The lead-atom electron configuration is $6s^2 6p^2$. The relativistic stabilization of the $6s$ shell (5, figure 11) is expected to raise the energy of Pb(IV) compounds, such as PbO_2 . This in turn is expected to explain much of the voltage of the lead-acid-battery reaction,



but it was unknown how much until the recent calculation by Ahuja et al. (80). These authors treated the solids Pb , PbO_2 , and PbSO_4 with and without relativity using two independent DFT codes. The electrolyte involves only light elements, and its ΔG was taken from experimental data. Four independent calculations found that the experimental electromotive force of +2.107 V was well reproduced by the average relativistic value of +2.13 V. The average nonrelativistic value was only +0.39 V. Hence cars start because of relativity.

The relativistic shifts in the energies of formation are shown in **Figure 4**. Not only does the reactant PbO_2 go up, but the product PbSO_4 of the discharge reaction (Equation 14) goes down. No clear interpretation exists for the latter trend.

Are other batteries strongly influenced by relativistic effects? For the mercury battery reaction



we find that 30% of the +1.35 V cell electromotive force comes from relativistic effects at the DFT level used (81). The strongest origin again was the relativistic destabilization of the Hg(II)O in which $6s$ electrons have been formally oxidized.

4.2. Shapes of Gold Clusters

Fairly comprehensive summaries on the theoretical chemistry of gold have been provided by Pyykkö (69, and references therein) and Schwerdtfeger & Lein (82). Work on gold clusters has been covered by Bonačić-Koutecký et al. (83), Garzón (84), Remacle & Kryachko (85, 86), Häkkinen (87), Johansson et al. (88), and Schooss et al. (89).

A particular issue is the molecular structure assumed by the neutral or charged gold clusters Au_n^q , $q = -1, 0, +1$ (see 69, table 7). A broad answer is that planar (2D) structures are preferred up to approximately an n of 11, 10, and 7 for these three charges, respectively. For higher n , three dimensions are preferred. The energy differences can be small, and the answer (a particular 2D or 3D structure) can depend on the theoretical method used. For the choice of DFT functionals, readers are referred to References 88 and 90. In WFT treatments of neutral Au_8 , one has to resort to large-basis CCSD(T) calculations to make it planar (D_{4h}) (91, 92). MP2 favors three dimensions. Experimentally, there is evidence for neutral 2D (C_s , not D_{3h}) Au_7 (93), but there is no information for its next neighbors. For anions, experiments favor 2D Au_{11}^- . For Au_{12}^- , there is evidence for both 2D and 3D isomers (88, 89). With regard to cations, Gilb et al. (94) measured a 2D D_{6h} Au_7^+ but found 3D structures for higher n .

A general, qualitative conclusion is that relativistic effects help to make the smaller gold clusters flat (see 95). The qualitative explanation is a stronger $5d$ - $6s$ hybridization, narrowed down to the doughnut-like $5d_{zz}$ - $6s$ orbital by Fernández et al. (96). As the silver $4d$ - $5s$ gap is larger, and hybridization is weaker, it then is logical that a silver substitution makes the $2D \rightarrow 3D$ transition arrive earlier (96, 97) than that for gold.

Quantum molecular dynamic studies suggest that the tendency to planarity may extend to the liquid phase for Au_n^- , $n = 11$ –14 (see 98). No experimental evidence exists for these relativistic flat liquids.

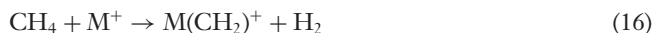
As mentioned above, a simple scalar relativistic explanation for the planarity is the easier $5d$ - $6s$ hybridization in the relativistic case. To the contrary, the SO favors three dimensions for anions around Au_{12}^- (88).

The larger, naked M_{55}^- clusters also show a difference. They are all approximately spherical but are of high symmetry (icosahedral) for $M = \text{Cu}$, Ag , and $\text{Au}(\text{NR})$, and of low symmetry for $M = \text{Au}(\text{R})$ according to calculations (99, 100) supported by photoelectron spectra.

Au_{58} has a major shell closing, but remains low symmetry, albeit almost spherical (101). This has been related to a known relativistic surface reconstruction, shrinking the $\text{Au}(100)$ surface area by 20%. For Au_{55}^- , the surface Au-Au distances shrank from 291 pm for I_h symmetry to 283 pm. Up to Au_{64}^- , the optimal structures build on the $n = 58$ one (101). The different individual coinage metals yield different cluster structures, up to very large n such as 40,000, obviously treated using fitted semiempirical potentials (102).

4.3. Platinum and Gold Catalysis

The gas-phase processes, typically studied by mass spectroscopy, have been reviewed by Schwarz (103). A notable example is the catalytic methane activation by Pt^+ . A driving factor for the reaction



is the bonding energy of the metal carbene $\text{M}(\text{CH}_2)^+$, which is 76, 68, and 112 kcal mol⁻¹ for $M = \text{Ni}$, Pd , and Pt , respectively. The relativistic origin of the large value for the $5d$ metal Pt was confirmed by a four-component Dirac calculation (104) (for other reactions of the carbene, see 103). Schwarz noted that the further, relativistically driven catalytic reactions include C-C

couplings, selective multiple C-F bond activations, alkene oxidations, and alkadiene oligomerizations. Another aspect is the spin-forbiddenness of ion-molecule reactions (105) (see also Section 4.4). The latest study on the bonding trends of the $M(\text{CH}_2)^+$ is Reference 106. Roithová & Schröder (107) explored the gas-phase chemistry of the coinage metals, whereas Benitez et al. (108) discussed the specific case of Au(I) carbenes and the $\sigma + \pi$ bonding of their reaction intermediates.

The homogeneous catalysis by Au(I) species in liquids was reviewed by Gorin & Toste (109). They noted that some driving forces behind the reactions are the strong Lewis acidity of both Au(I) and Au(III), the occasional aurophilic attraction between two or more Au(I)s, the strengthening of Au-L bonds, the tendency of Au(I) to two coordination (eliminating further ligands easily), and the above-mentioned stability of the carbenoids, all of which can be related to relativistic mechanisms. For more experimentally oriented reviews, readers are referred to References 110–112.

Another vast area is the catalysis by gold nanoparticles, including the treatise by Bond et al. (113) and reviews by Ishida & Haruta (114), Chen & Goodman (115), and Hutchings (116) (for individual examples, see also 69). Typical questions in theoretical work concern the role of support effects and charging of the nanocluster on surfaces, geometric fluxionality, size dependence, heights of the reaction barrier, and the HOMO-LUMO energy gap (117, 118). Oxygen vacancies on the oxide substrate may be important (119). Special gold sites of the cluster may be essential, such as ones with a low coordination number (120) or ones in the perimeter of the nanoparticle-substrate interface (121). Little information is available on the explicit role of relativity or on systematic silver/gold comparisons, but we mention gold nanocatalysis because of its importance.

4.4. Spin-Forbidden Chemical Reactions

Conical intersections have been reviewed, for example, by Matsika & Yarkony (122), Domcke et al. (123), and especially Poluyanov & Domcke (124). Tatchen et al. (125) presented an example on psoralen (125), whereas Schröder et al. (126), Poli & Harvey (127), and Gülich & Goodwin (128) presented models of inorganic and organometallic reactions.

4.5. Polonium

A striking example is the simple cubic structure of polonium. Without relativistic effects, polonium would have the same structure as tellurium. With relativistic effects, the correct structures of α - and β -Po could be reproduced. Legut et al. (129) and Verstraete (130) have presented the latest results, and major effects have been found on the elastic constants. Free-energy calculations were added by Verstraete.

4.6. Spin-Orbit Effects in Structural Chemistry

Many relativistic effects on chemistry could already be seen at the scalar-relativistic (SO-averaged) level. These effects were typically related to the energetic stabilization of the *s* and *p* shells, and/or the destabilization of the *d* and *f* shells.

4.6.1. Molecular groups. A recent, rare example of an SO-induced change of molecular structure is the octahedral $[\text{Ti}_6]^{6-}$ polyanion in solid $\text{Cs}_4\text{Ti}_2\text{O}$, synthesized in Jansen's group (131). With two electrons more (26e), the Wade rules would predict an octahedral structure. With 24 valence electrons only, they would predict a Jahn-Teller (JT) distortion. A relativistic band-structure calculation, including SO, opens up a gap at the Fermi level and prevents the JT distortion. A

similar SO versus JT competition was found earlier for the $[\text{Ti}_8]^{6-}$ clusters in solid $\text{Cs}_{18}\text{Ti}_8\text{O}_6$ (132). Both solids are diamagnetic.

Similar symmetry changes were found computationally for the free Pb_n clusters (133). With SO effects, the $n = 6$ case is an O_h octahedron. Without SO effects, a D_{4h} structure is found instead.

Among transition metal clusters, SO effects help to make Pt_n clusters ($n = 3-5$) flat (134), an effect opposite of that for the Au_n^- anions (88).

An early, experimentally observed, SO-induced system is the Bi(I) compound $\text{Bi}^+(\text{Bi}_9^{5+})(\text{HfCl}_6^{2-})_3$ of Friedman & Corbett (135). For some further examples on JT distortions cancelled by SO effects, see Dylla & Faegri (32, p. 467) or David et al. (151).

Finally, there are large SO effects on centered compounds EL_n . Whereas the rare-gas tetrafluorides $(\text{Rg})\text{F}_4$ remain D_{4h} for $\text{Rg} = \text{Xe}$ and Rn , the case $\text{Rg} = \text{E118}$ is calculated to have another, SO-induced, higher-symmetry T_d minimum that is slightly lower (136, 137). Moreover, the next s orbital, the $8s$, is calculated to come in, giving an example on a *pre- s* element, using a term from Reference 6. The $(\text{E117})\text{F}_3$ is calculated to adopt the higher-symmetry D_{3h} structure, both with and without SO effects (138). The C_{2v} minimum, typical for light-halogen XY_3 systems, exists only at the nonrelativistic level.

These are changes of molecular symmetry. Concerning SO effects on the molecular binding energy, starting with diatomics, the system can dissociate to atoms with some open heavy-atom $l > 0$ shells or to no such shells, cases A or B, respectively (139, see chapter 1V.A.8 and figure 38). Here BiH^+ , Pb_2 , PbO , PbH and Ti_2 are examples of case A, whereas Au_2 , AuH , CsHg , and noble-gas dimers are examples of case B. In case A, the larger SO stabilization of the atom, compared with that of the molecule, decreases the dissociation energy, D_0 . In case B, there is no such first-order SO change. However, a second-order SO effect can increase D_0 . For further case A examples, readers are referred to Reference 9.

4.6.2. Solids. SO effects substantially contribute to the phonon spectrum and specific heats, $C(T)$, of Bi (140), Pb (141), and PbX ($X = \text{S}, \text{Se}, \text{Te}$) (142). In Bi, a slight scaling of the SO interaction gave a $C(T)$ in perfect agreement with experiment. The SO increased the lattice parameter, a_0 , and substantially decreased the cohesive energy (140, 143). In Pb (141), SO effects also helped to reproduce the phonon spectra. In eka-lead, E114, the SO removes most of E_{coh} , leaving only 0.5 eV per atom (143). It is almost a noble gas, as anticipated by Pitzer (3). Indeed, the E114 atom has no EA (144). For lead, the trend is already there, but weaker.

4.7. Topological Insulators and Graphene

New, essentially relativistic phenomena are the topological insulators in which an insulating compound, such as $\text{Bi}_x\text{Sb}_{1-x}$, or Bi_2Se_3 (145), has metallic surface states, created by SO coupling (for reviews, see 146 and 147). Similar SO effects may arise in the Mott insulator transition (148).

In this context we also note the apparently relativistic behavior of graphene, whose linear dispersion relation (149, see equation 1.2) is of the same type as that for a free Dirac particle:

$$E_{\pm}(q) = \hbar v_F q + O(q/k)^2. \quad (17)$$

Here E is the energy, q is the wave vector, and the equivalent of c is $v_F = 3ta/2$, where a is the lattice parameter and $t \approx 2.5$ eV is a hopping frequency. Hence the relation in Equation 17 has nothing to do with relativistic effects, although its form is similar.

SUMMARY POINTS

1. The classical examples of relativistic effects in chemistry remain and have been included in most chemistry textbooks.
2. One of the oldest examples, which deserves more attention, is the SO-induced NMR heavy-atom shift.
3. Investigators continue to discover new examples, such as the heavy-element batteries.
4. Catalysis is one of the most important applications of relativistic quantum chemistry.
5. The SO effects in structural chemistry have been identified only recently after technical progress.

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